

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

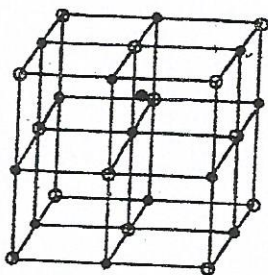
13

MARKING SCHEME

JUNE 2013

CHEMISTRY 9189/1

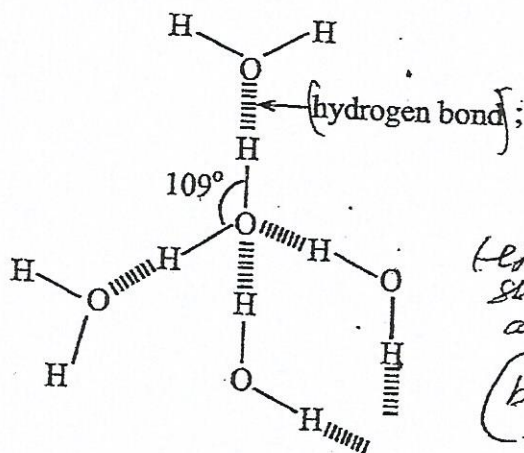
(a) magnesium oxide

○ = Mg^{2+} ● = O^{2-}

} clear labelling

Accept correct
labelling

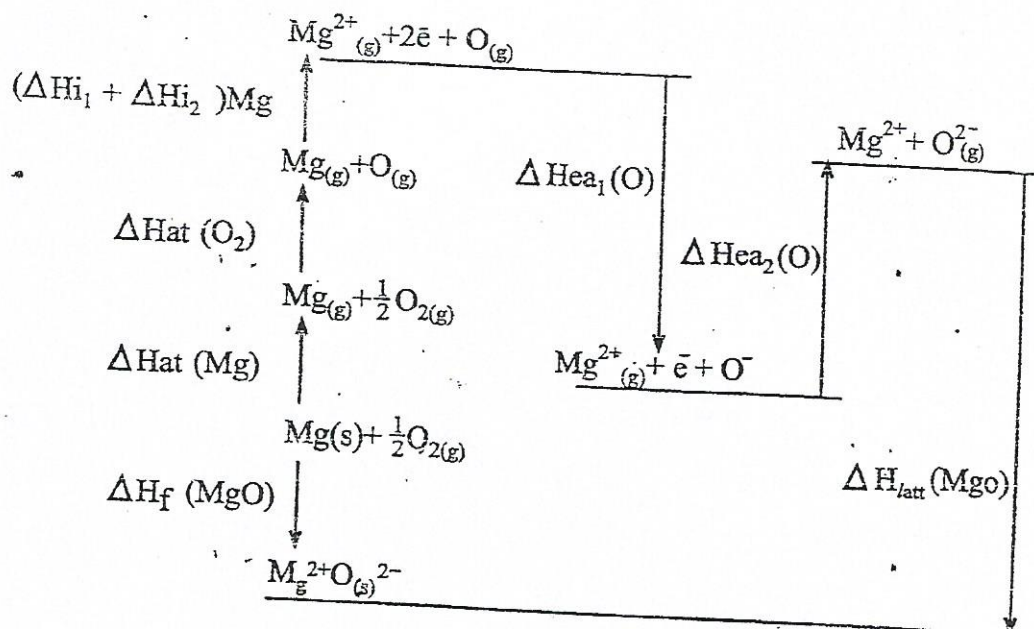
ice

tetrahedral
structure
around O atom
(bond angle 109°)
+ H₂ bonds

[2]

- (b) (i) High charge densities of small Mg^{2+} and O^{2-} ions result in strong ionic bonds (which cannot be easily broken by polar water molecules) / AW: Strong ionic bonds [1]
(A comparison of ionic bonds $\text{Mg}(\text{OH})_2$ in MgO)
- (ii) hydrogen bonds between water molecules in ice are weaker than ionic bonds in MgO / AW; [1]
- (iii) Extensive hydrogen bonding gives open tetrahedral structure in ice / larger mass per unit volume in liquid water where hydrogen bonds are fewer and particles closely packed; / AW [1]
(comparison must be clear)
- (iv) Attraction is stronger between smaller Mg^{2+} and O^{2-} ions with larger charge densities than larger Na^+ and Cl^- ions, / AW [1]

(c) (i)



Correct cycle; + balanced eqn [1]

Correct labels; [1]

State symbols [1]

$$(\Delta H_{\text{i}1} + \Delta H_{\text{i}2})\text{Mg} = +736 + 1450 = +2186$$

$$\Delta H_{\text{at}} \text{ O} = \frac{+496}{2} = +248$$

[1]

$$\Delta H_{\text{f}}(\text{MgO}) = +150 + 248 + 736 + 1450 - 142 + 775 - 3933 = -716 \text{ kJ mol}^{-1}$$

[1]

[Total 12]

(iii) high repulsion between similarly charged species/electron vs negatively charged O^{2-} / AW

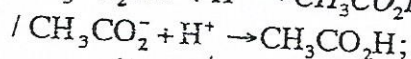
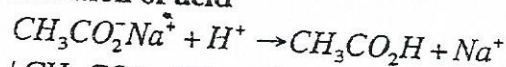
[TOTAL 12]

(a) proton donor; [1]

(b) (i) Highly electronegative Cl atoms draw electrons towards themselves thereby weakening the O-H bond/stabilising the $\text{CCl}_3\text{CO}_2^-$ ion / AW; [1]Electron donating CH_3 gp strengthens the O-H bond in $\text{CH}_3\text{CO}_2\text{H}$ / destabilises the CH_3CO_2^- ion / AW; [1](ii) Higher charge density of Al^{3+} result in more polarisation of water molecules to release H^+ ions than with larger Mg^{2+} ion / AW; [1]

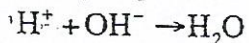
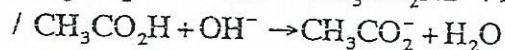
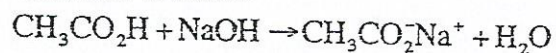
(c) (i) Resulting solution consists of a mixture of a weak acid and its salt; [1]

Addition of acid



[1]

Addition of alkali



together

[1]

$$(ii) \quad [\text{CH}_3\text{CO}_2\text{H}] = 0,1 \times \frac{(25-15)}{25+15} = 0,025 \text{ mol dm}^{-3}$$

[1]

$$[\text{CH}_3\text{CO}_2\text{Na}^+] = 0,1 \times \frac{15}{(25+15)} = 0,0375 \text{ mol dm}^{-3}$$

[1]

$$\text{pH of buffer} = \text{pK}_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$

*After value
calculation
accepted*

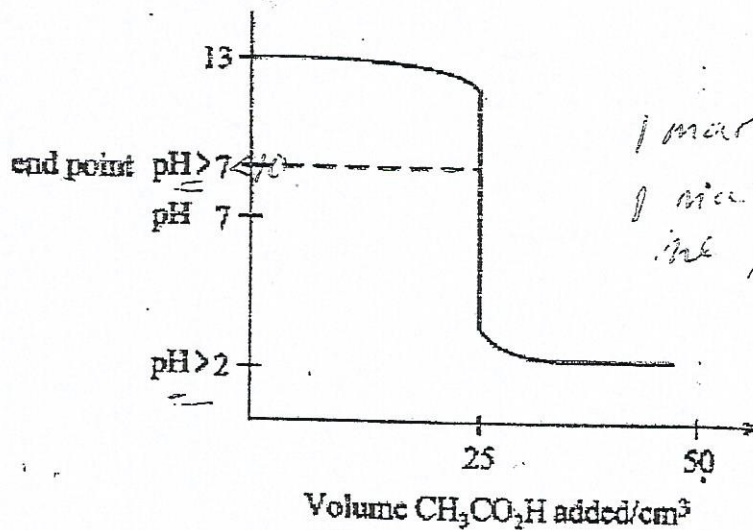
$$= 4,76 + \log \frac{0,0375}{0,025}$$

[1]

$$= 4,94$$

[1]

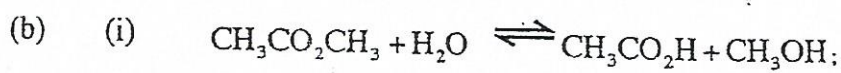
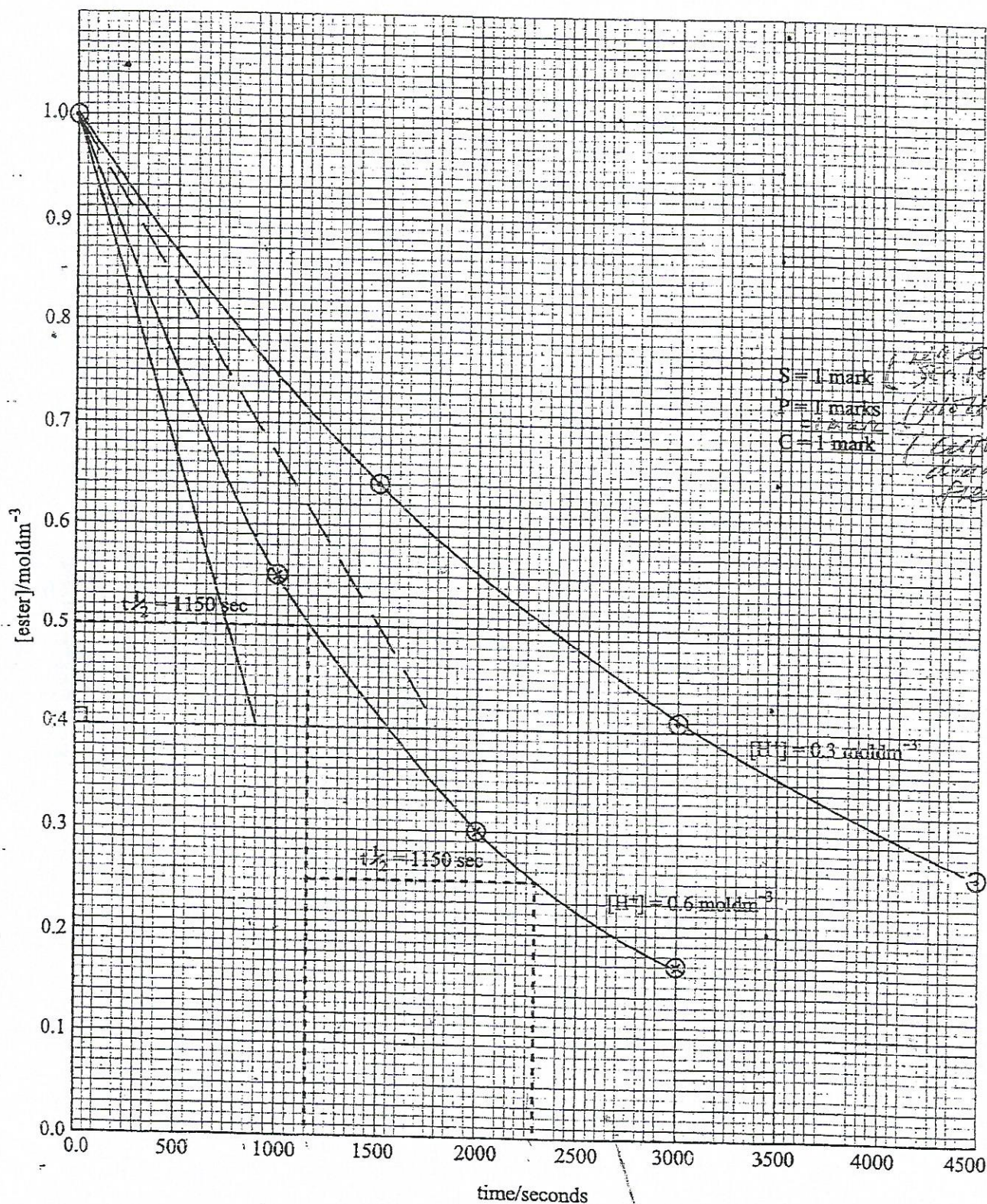
(iii)



*1 mark correct shape
1 mark for correct labels
the pHs and volumes*

[Total: 12]

- 3 (a) Time taken for concentration of a reactant to reduce to half of its original value/AW; [1]



(ii) On graph paper;

[4] [4]

- (iii) order wrt [ester] = 1;
since $t_{\frac{1}{2}} = 1150$ sec is constant [1]

(shown on graph) / (1100-1200 sec); [1]

- (iv) initial rate when $[H^+]$ is 0.3 = $\frac{0.6}{1800}$;
(tangent shown on graph) = $3.33 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$ [1]

Initial rate when $[H^+]$ is 0.6 = $\frac{0.6}{900}$ [1]

$\therefore$ order wrt $[H^+] = 1$ since initial rate doubles when $[H^+]$ is doubled; [1]

- (v) Rate = $K[\text{ester}][H^+]$; [1]

[Total: 12]

- 4 (a) $\text{Cu} + 2\text{H}_2\text{SO}_4 \rightarrow \text{CuSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$; [1]

- (b) (i) colour of KMnO_4 changes from purple to pink / colourless; [1]

- (ii) $2\text{MnO}_4^- + 5\text{SO}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{Mn}^{2+} + 5\text{SO}_4^{2-} + 4\text{H}^+$; [1]
 $E^\circ_{\text{cell}} = +1.35 \text{ V}$ [1]

- (c) (i) $\text{NO}(\text{g}) + \frac{1}{2}\text{O}_2 \rightarrow \text{NO}_2(\text{g})$; [1]
 $\text{NO}_2(\text{g}) + \text{SO}_2(\text{g}) \rightarrow \text{SO}_3(\text{g}) + \text{NO}(\text{g})$; [1]

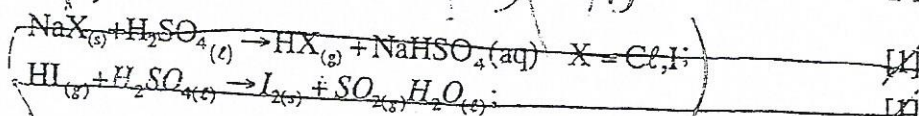
($\text{SO}_3(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{SO}_4(\text{aq})$ acid rain;)

- (ii) Destroys limestone buildings / corrugated iron roof sheets/
steel structures/vegetation/aquatic life/causes soil to be acidic; [1]

- (d) (i) $\text{SO}_3(\text{g}) + \text{H}_2\text{SO}_4(\text{l}) \rightarrow \text{H}_2\text{S}_2\text{O}_7(\text{l})$; [1]

$\text{H}_2\text{S}_2\text{O}_7(\text{l}) + \text{H}_2\text{O}(\text{l}) \rightarrow 2\text{H}_2\text{SO}_4(\text{l})$; [1]

- (ii) white fumes (of HCl) with NaCl ; black ($\text{I}_{2(\text{s})}$) / purple (I_2) vapour [1]
with NaI [1]

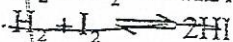
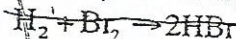


[Total: 12]

- 5 (a) (i) Br_2 reacts with H_2 only on heating (300°C) and in the presence
of a (platinum) catalyst; [1]

I_2 combines with H_2 only partially and slowly even on heating [1]

(300°C) and in the presence of a catalyst;

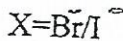
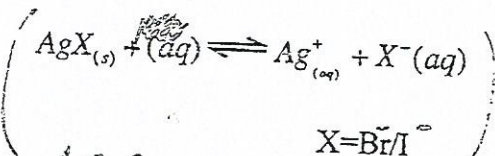


$\text{H}_2 + \text{Br}_2 \rightarrow 2\text{HBr} + \text{cond.}$
 $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI} + \text{cond.}$
 HI is more stable than HBr
 without conditions

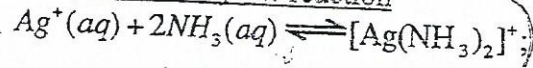
H-I bond, (B E 299 KJ mol⁻¹) is much weaker than H-Br bond
 (B E 366 KJ mol⁻¹) $\Delta H_f \text{ HI} = +26 \text{ KJ mol}^{-1}$,
 $\Delta H_f \text{ HBr} = -36 \text{ KJ mol}^{-1}$;

HI is energetically less stable than the reactants whilst
 HBr is energetically more stable than the reactants.

- (ii) Following equilibria are established on adding $\text{NH}_3(\text{aq})$

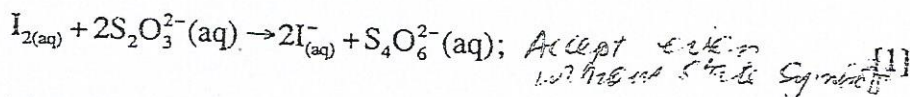
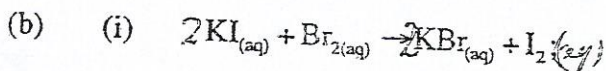


$[\text{Ag}^+]$ lowered by the reaction



$\text{AgBr}_{(s)}$ (partially dissolves) because ionic product is almost equal
 but slightly greater than K_{sp} ; / AW

$\text{AgI}_{(s)}$ (insoluble) because K_{sp} value is very low and ionic product
 still exceeds K_{sp} despite lowering of $[\text{Ag}^+]$ / AW;

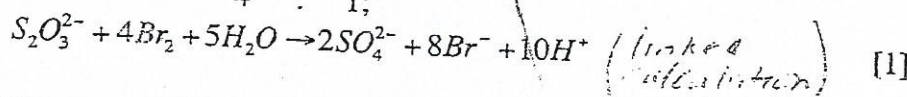


(ii) Mole ratio $\text{I}_2 : \text{S}_2\text{O}_3^{2-} = 1:2$

$$\therefore n(\text{I}_2) = \frac{1}{2} n(\text{S}_2\text{O}_3^{2-}) = \frac{1}{2} \times \frac{19.5}{1000} \times 0.120 = 0.00117;$$

$$(iii) n(\text{S}_2\text{O}_3^{2-}) = \frac{2.45}{1000} \times 0.120 = 0.000294;$$

Mole ratio $\text{Br}_2 : \text{S}_2\text{O}_3^{2-}$
 0.00117 : 0.000294
 4 : 1;



(c) Br_2 is a stronger oxidising agent than iodine;

Br_2 oxidises sulphur from +2 state in $\text{S}_2\text{O}_3^{2-}$ to +6 in SO_4^{2-}
 whereas I_2 oxidises it to +2 $\frac{1}{2}$ in $\text{S}_4\text{O}_6^{2-}$;

(b) specification of oxidation
 numbers.

[Total: 12]

6

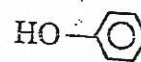
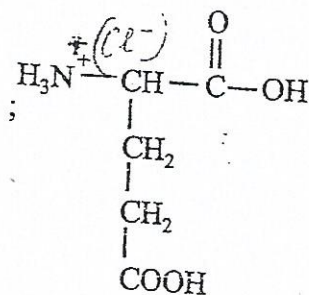
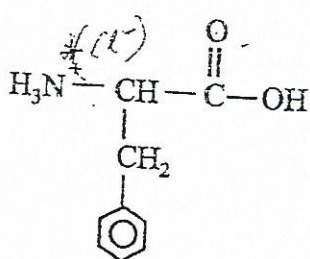
(a) (i)

amine / amino; ester; amide; carboxyl; phenyl

(1 mark for any correct 2)

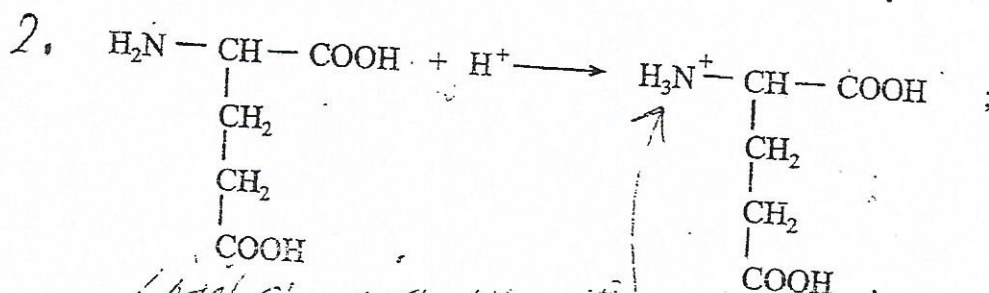
first four. provided these are in [2]

(ii)

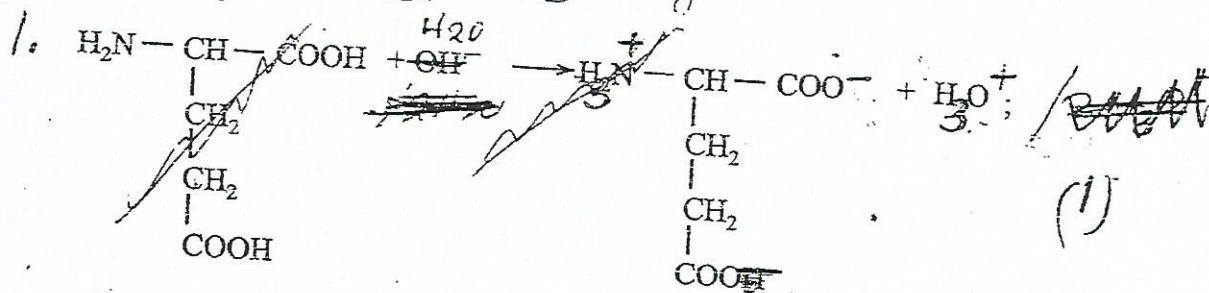


[3]

(b) on adding acid



AW showing 1st 1st NH₂ group with H⁺ [1]
on adding base - water

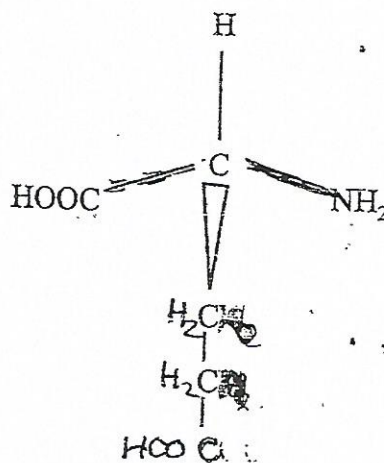
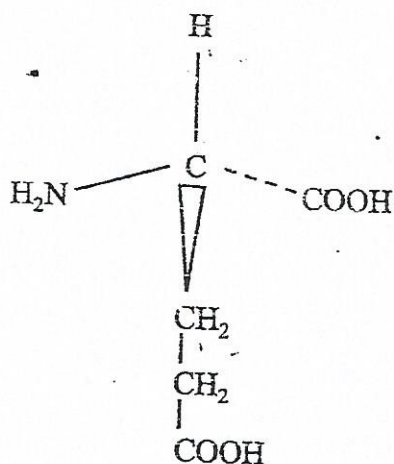


(1)

(ii) optical;

[1]

(iii)



* correct reflector (1)
+ tetrahedral shape

* isomers + mirror (2)
* isomer no mirror (1)

* 4 different
bonds
are wrong
because the
are on
same plane

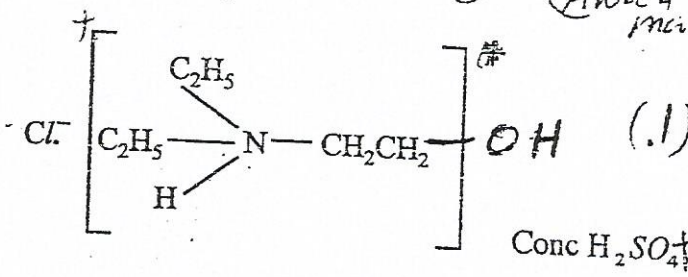
- (c) exists in the form of a zwitterion; [1]
ionic charges increase the attractive forces between molecules in the solid; [1]
/AW [Total: 12]

7 (a) soluble; because it is ionic; [2]

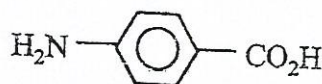
(b) (i) Step Reagent

Conditions

- I Sn, conc HCl (1) reflux; boil (1) ~~boil~~
II ~~KMnO₄~~ / K₂Cr₂O₇ (aq) ; H₂SO₄ (aq), Reflux/boil (2)
III (1) ~~(K⁺ H⁺)~~ ~~Award Substage mark~~



(ii)



- (iii) Step Type of reaction [1]
I reduction; [1]
II oxidation; [1]
III esterification; / acid chloro elimination [1]
[Total: 12]

8

(a) **Addition Polymerisation**

many small unsaturated molecules (monomers) combine to form a single product which is a macromolecule called a polymer / AW;

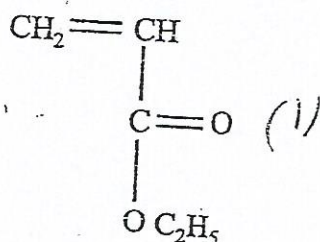
[1]

Condensation Polymerisation.

many small molecules (monomers) combine to form a macromolecule, the reaction being accompanied by the elimination of a small molecule (such as water or hydrogen chloride) / AW;

[1]

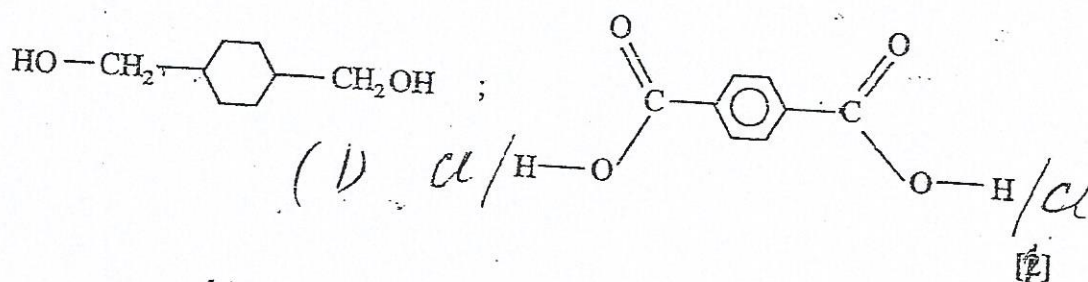
(b) (i)



; Addition polymerisation (1)

[2]

(ii)



[2]

condensation polymerisation;

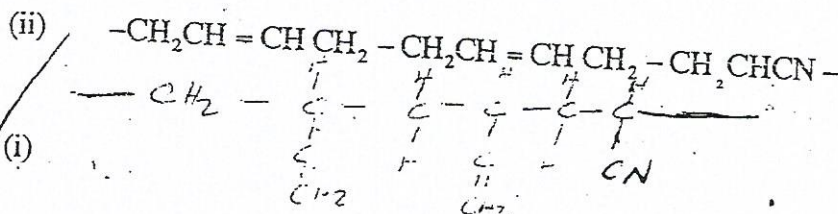
[1]

(c) (i) A 1,3 butadiene / buta-1,3-diene;

[1]

B propenenitrile;

[1]



[1]

(d)

(i)

